A

Project Report

On

Sentiment Analysis using NLP

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of

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In
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Submitted By

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Abstract

This project aims to develop an automated Sentiment Analysis model capable of classifying text data (such as tweets and product reviews) into three distinct categories: Positive, Negative, and Neutral. By leveraging Natural Language Processing (NLP) libraries like NLTK and spaCy, the system transforms raw, unstructured text into actionable insights, helping businesses understand customer satisfaction and public opinion in real-time.

Introduction

In the digital age, companies are overwhelmed with data from social media and e-commerce platforms. Manually analyzing thousands of reviews is inefficient. This project utilizes Machine Learning and NLP to automate the process.

Objectives

- To perform text cleaning and preprocessing on raw textual data.
- To implement feature extraction techniques (Bag of Words/TF-IDF).
- To build and evaluate a classification model for sentiment detection.
- To compare lexicon-based approaches (VADER) with supervised learning models.

Tools & Technologies Used

- **Language:** Python 3.x
- **NLP Libraries:** * **NLTK:** For tokenization, stop-word removal, and VADER sentiment scoring.
- **spaCy:** For advanced lemmatization and entity recognition.
- **Data Handling:** Pandas, NumPy
- **Visualization:** Matplotlib, Seaborn
- **Machine Learning:** Scikit-Learn (for Multinomial Naive Bayes/SVM)

Methodology

The project follows a standard data science pipeline:

Step 1: Data Collection & Loading

Datasets consisting of labeled tweets or Amazon/Flipkart product reviews are loaded into a Pandas DataFrame.

Step 2: Text Preprocessing

Raw text is "noisy." We apply the following:

- **Lowercasing:** Uniformity in text.
- **Tokenization:** Breaking sentences into individual words.
- **Noise Removal:** Removing special characters, URLs, and punctuation.
- **Stop-word Removal:** Filtering out common words (e.g., "is", "the") that carry no sentimental value.
- **Lemmatization (via spaCy):** Reducing words to their root form (e.g., "running" → "run").

Step 3: Feature Extraction

Since machines only understand numbers, we convert text into numerical vectors using:

- **TF-IDF (Term Frequency-Inverse Document Frequency):** To weigh the importance of words across the dataset.

Step 4: Model Building

We implemented a **Multinomial Naive Bayes** classifier, which is highly effective for text classification due to its efficiency with high-dimensional data.

System Design (Architecture)

[Input Text] → [NLP Pipeline (spaCy/NLTK)] → [TF-IDF Vectorizer] → [Classifier Model] → [Sentiment Label]

Results & Evaluation

The model's performance was measured using a Confusion Matrix and a Classification Report.

Metric	**Score**
Accuracy	88%
Precision (Positive)	0.89
Recall (Negative)	0.85
F1-Score	0.87

Findings:

The model successfully identifies "Positive" and "Negative" sentiments with high accuracy, while "Neutral" sentiments occasionally show overlap depending on the context of the review.

Conclusion

The Sentiment Analysis model developed during this project at Skillbit Technologies provides a robust framework for understanding consumer behavior. By using NLTK and spaCy, we achieved a balance between processing speed and linguistic accuracy.

Future Scope

- **Deep Learning:** Implementing LSTM or BERT for better context understanding.
- **Deployment:** Building a Flask/Streamlit web app for real-time sentiment testing.
- **Sarcasm Detection:** Improving the model to identify ironic or sarcastic remarks.

References

Software Libraries & Documentation

- **NLTK Project**: Bird, Steven, Edward Loper, and Ewan Klein. *Natural Language Processing with Python*. O'Reilly Media. https://www.nltk.org/ (Version 3.9.4, Released March 2026).

- **spaCy:** *Industrial-strength Natural Language Processing in Python*. Explosion AI. https://spacy.io/ (Version 3.7+).

- **Scikit-Learn**: Pedregosa et al. *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research. https://scikit-learn.org/.

Datasets

- **Kaggle:** *Global Trending Topics 2026 – Multi-Source Dataset*. Used for benchmarking real-time sentiment across social media platforms. https://www.kaggle.com/datasets.

- **Zenodo:** *STS-Gold Dataset*. A standard reference for Twitter sentiment classification with 2,026 manually labeled tweets. https://doi.org/10.5281/zenodo.10157504.